

RESEARCH INTEGRITY TOOLKIT

A Framework for Detection, Evidence & Remediation

Organised by unit of analysis: Individual → Team → Group / Collusion → Field / Community

Tier 1

Individual level

Tier 2a

Team level

Tier 2b

Group / collusion

Tier 3

Field / community

4 slides per problem row · Introduction · Evidence of Scope · Detection Tools · Prevention & Remediation

TIER 1

**Individual-level
misconduct & errors**

DATA FABRICATION

A single researcher invents raw data — often a junior team member or PI under grant and publication pressure. The most severe and clear-cut form of research misconduct, it spans a spectrum from cherry-picked exclusions to wholly invented datasets.

SCALE OF PROBLEM

~67% of medical retractions

Misconduct — fabrication, falsification, plagiarism — accounts for the majority of retractions in biomedical research.
(Fang et al. 2012, PNAS)



EVIDENCE OF SCOPE

01 ORI annual reports (1992–)

The US Office of Research Integrity averages ~10 misconduct findings per year in federally funded research, though capacity has declined sharply — only 2 findings in 2025.

03 Stapel case (2011)

Dutch social psychologist Diederik Stapel fabricated data in dozens of studies over ~10 years. 58 papers retracted; doctoral students uncovered the fraud.

02 Fang et al. (2012, PNAS)

Misconduct — fabrication, falsification, and plagiarism — accounts for ~67% of retracted medical papers; only ~21% due to honest error.

04 Gino case (Harvard 2024)

Investigative committee found Francesca Gino 'committed research misconduct intentionally' — 168 participant responses altered across 4 studies.

DETECTION TOOLS

01	GRIM test (2016)	Flags impossible reported means given integer data and sample size. Brown & Heathers: if N=20 integer responses, only specific decimal means are possible.
02	SPRITE (2018)	Reverse-engineers plausible raw datasets from summary statistics. Revealed one Wansink study implied a child ate ~60 carrots in a single sitting.
03	statcheck	Automated extraction and recalculation of p-values from PDFs; flags decision-changing inconsistencies.
04	Image forensics / Bik	Visual and AI-assisted detection of manipulated figures — western blots, microscopy, gels. Bik has led to 1,331+ retractions since 2014.
05	ORI investigation (1992–)	Federal investigators review institutional reports for research involving NIH, CDC, and other HHS agencies.

TIER 1 — INDIVIDUAL LEVEL

DATA FABRICATION



INTRO



EVIDENCE



DETECTION



PREVENTION

PREVENTION & REMEDIATION

■ Open raw data sharing

Enables auditing by any reader. Increasingly required by journals and funders — raw data exposes impossible values immediately.

■ Lab notebooks & audit trails

Timestamped records of data collection; electronic lab notebooks create immutable provenance logs.

■ Bik Science Integrity Fund (2025)

Financial and legal support for volunteer sleuths. Launched in partnership with Retraction Watch / Center for Scientific Integrity.

■ Data DOIs & versioning

Zenodo (2013), Figshare (2012), DataCite (2009–): persistent identifiers create a citable, versioned audit trail for datasets.

■ Federal whistleblower protections

Legal protection for researchers who report misconduct by colleagues; Research Integrity Officer role at institutions.

IMAGE MANIPULATION

Duplicated, spliced, rotated, or AI-generated figures — western blots, microscopy images, flow cytometry, gel images. The most common single route into detection of fraud. Increasingly hard to catch as generative AI matures.

PREVALENCE

**3.8% of papers
had problematic figures**

Among 20,000+ papers screened 1995–2014;
roughly half showed signs of deliberate
manipulation.

(Bik et al. 2016, mBio)



EVIDENCE OF SCOPE

01 **Bik et al. (2016, mBio)**

3.8% of 20,000+ papers published 1995–2014 had problematic figures; ~50% of those showed signs of deliberate manipulation, not accidental duplication.

03 **Tessier-Lavigne (Stanford 2023)**

Stanford president resigned after 8-month investigation into image manipulation flagged by Elisabeth Bik and others on PubPeer.

02 **Imagetwin / J. Cancer (2024–25)**

AI scanning found 18–20% of papers in one journal contained problem images — before and after the journal changed its correction-charge policy.

04 **Dana-Farber (2024)**

Data sleuth David Sholto identified image problems across multiple papers from the Harvard-affiliated cancer institute; several retractions followed.

DETECTION TOOLS

01	Bik visual screening (2014–)	Systematic manual scanning of published papers for duplicated or manipulated figures. By 2024, Bik's work has led to 1,331 retractions and 215 expressions of concern.
02	Imagetwin AI (early 2020s)	Automated batch scanning of submitted PDFs for duplicated or manipulated images; deployed by publishers at submission stage.
03	Proofig AI (early 2020s)	Publisher-grade image integrity checking; analyses pixel-level similarities to flag potential manipulation before peer review.
04	PubPeer crowdsourcing (2012–)	Open post-publication annotation platform enabling global volunteer reviewers to flag concerns; regularly precedes formal institutional action.



PREVENTION & REMEDIATION

■ Raw image submission at review

Requiring unprocessed original image files alongside manuscript — any discrepancy between raw and published is immediately detectable.

■ COPE retraction guidelines (2009–)

Committee on Publication Ethics standardised criteria for when and how publishers should issue corrections, expressions of concern, and retractions.

■ Figure metadata standards

Embedding provenance in image files; some journals now require checksums and acquisition metadata to accompany submitted figures.

■ Publisher AI screening (2023–)

Wiley, Springer Nature, and Elsevier now deploy automated image-integrity checking at the submission stage, before peer review assignment.

PLAGIARISM & TEXT RECYCLING

Copying text or results from other works without attribution; self-plagiarism across multiple papers from the same dataset; AI-generated boilerplate that may not match any specific source. Detection tools are now in an arms race with AI paraphraser.

EMERGING THREAT

14% of biomedical abstracts in 2024

Showed markers of AI-generated text, per Nature news report (July 2025). AI-generated text is technically distinct from plagiarism but raises similar integrity concerns.

EVIDENCE OF SCOPE

01 Cabanac et al. (2021)

Identified 2,000+ papers using 'tortured phrases' — systematic obfuscation to evade detectors ('artificial neural network' becomes 'counterfeit brain organisation').

03 Problematic Paper Screener

Over 32,786 suspected paper-mill products identified partly via tortured-phrases signatures — obfuscation is a hallmark of purchased manuscripts.

02 Nature (July 2025)

14% of biomedical abstracts published in 2024 showed probabilistic markers of AI-generated text; proportion rising sharply.

04 CrossCheck / iThenticate data

Publishers report dramatic increase in manuscripts flagged for similarity since 2022; much of the increase attributed to AI paraphrasing rather than direct copying.

DETECTION TOOLS

01

iThenticate / CrossCheck (2008–)

Text similarity matching against published literature at submission. Mandatory at most major publishers; owned by Turnitin.

02

Tortured phrases screener (2021)

Flags tell-tale obfuscation patterns from machine translation — statistical signatures of automated paraphrasing.

03

AI-text detectors (2023–)

Tools like GPTZero and Originality.ai detect probabilistic signatures of LLM output; reliability concerns remain, especially after paraphrasing.

PREVENTION & REMEDIATION

■ Mandatory plagiarism check

Standard submission requirement at most major publishers; typically uses iThenticate / CrossCheck before peer review assignment.

■ AI disclosure requirements (2023–)

Authors must declare AI use in drafting at most major journals; Nature, Science, Lancet, and hundreds of others have adopted policies.

■ COPE self-plagiarism policies

Clear guidance on acceptable text reuse thresholds; most journals require disclosure when reusing own prior published text.

■ Preprint deposits

Time-stamped preprints on arXiv, bioRxiv, SSRN establish priority and create a public record that makes post-hoc plagiarism detectable.

RESEARCH METHODS ERRORS

Coding bugs, data entry errors, miscoded variables, mathematical errors in proofs. Error rates scale super-linearly with project complexity — doubling lines of code more than doubles errors, because code functions as an interdependent network, not a linear proof.

THE CORE PROBLEM

Error rates scale super-linearly

A constant errors-per-line rate produces more than proportional impact as code networks grow. One upstream error cascades through all downstream components.

EVIDENCE OF SCOPE

01 Reinhart & Rogoff (2010 / Herndon et al. 2013)

Excel error excluded 5 countries, reversing the debt-growth conclusion that informed austerity policy across Europe and North America.

03 Brodeur et al. (IZA 2023)

Coding errors found in ~25% of economics papers when re-executed by independent teams using shared replication packages.

02 Eklund et al. (2016, PNAS)

fMRI software bug inflated false-positive rates — potentially affecting ~40,000 published neuroimaging studies over 15 years.

04 Chang & Li (2022)

Only 14–43% of published economics studies were computationally reproducible on first independent attempt.

DETECTION TOOLS

01	Re-execution checks	Independent run of published code on shared data; any divergence from reported numbers signals an error worth investigating.
02	AEA Data Editor (2018–)	American Economic Association requires code packages to execute cleanly before publication; adopted as a model across disciplines.
03	PLOS ONE reproducibility (~2014)	Editorial verification of code outputs at submission; first major journal to adopt systematic computational reproducibility checks.
04	Unit tests	Software engineering practice: automated checks confirm code components behave as expected after any change.
05	statcheck	Automated extraction and recalculation of p-values from PDFs; catches statistical reporting inconsistencies that may indicate computational error.

PREVENTION & REMEDIATION

■ Code sharing mandates

NIH & NSF (2023): federally funded research must deposit executable code alongside data at publication.

■ Software citation standards

FORCE11 (2016): code and data treated as citable outputs with persistent DOIs — creates an accountability trail for each version.

■ Gentzkow & Shapiro guide (2014)

Code and Data for the Social Sciences: canonical practitioner reference on automation, version control, and abstraction.

■ Containerised environments

Code Ocean / Binder (2017–): sealed compute capsules preserve exact software environment; results reproducible years later on any machine.

■ Unit testing norms

Automated regression tests catch downstream breaks when upstream components change; standard in software, increasingly adopted in research.

TIER 2A

**Team-level
analytic issues**

COLLABORATIVE PIPELINE FAILURE

Errors emerge from handoffs between team members working on interdependent data, code, and models. A change in one component cascades downstream. Most common when inexperienced members (RAs, grad students) build the pipeline without coordination systems. Described by Gentzkow & Shapiro (2014).

NARPS (2020, NATURE)

70 teams, same data

Divergent conclusions on 5 of 9 hypotheses. Teams using identical data reached opposite significant results — a direct consequence of analytic pipeline flexibility.

EVIDENCE OF SCOPE

01 NARPS (Botvinik-Nezer et al. 2020, Nature)

70 independent teams, same brain imaging data → divergent conclusions on 5 of 9 hypotheses. Same freedom of pipeline choice produced opposite significant results.

03 Brodeur et al. (IZA 2023)

Pipeline and environment failures common even in packages explicitly described as computationally reproducible by the submitting authors.

02 Gentzkow & Shapiro (2014)

Catalogue of real project failures: code calling moved files, inconsistent sample definitions across scripts, RA directories impossible to reverse-engineer months later.

04 AlphaChip controversy (2024, Comms ACM)

Data leakage, inconsistent environments, and undisclosed software versions; benchmark performance not reproducible by independent groups.

DETECTION TOOLS

01

Push-button reproducibility tests

Delete all outputs; re-run from raw data to final tables with a single command. Any deviation surfaces a handoff error.

02

Pull-request diff review

No direct commits to shared codebase without a peer approving the change; diff view makes downstream effects of any change explicit.

03

targets / drake (R, 2018–)

Declarative dependency graphs track which outputs are out of date when inputs change; only affected components are re-run.

04

Environment audit (renv / Docker)

Verify software stack — packages, versions, OS — matches exactly what was used in the analysis that produced the reported results.

05

Data manifest / provenance logs

Record all transformations applied to each data file; enables pinpointing exactly where a discrepancy was introduced.

TIER 2 A — TEAM LEVEL

COLLABORATIVE PIPELINE FAILURE



INTRO



EVIDENCE



DETECTION



PREVENTION

PREVENTION & REMEDIATION

■ Git / version control (~2012)

Every change to code or data is logged with author, timestamp, and message. Enables bisecting to find exactly when a bug was introduced.

■ targets / drake pipeline tools

R-ecosystem workflow managers (2018–): describe analysis as a directed graph; re-run only what is out of date after any change.

■ Code Ocean / Snakemake / Nextflow

Push-button reproducibility capsules (Code Ocean 2017–) and workflow managers (Snakemake / Nextflow 2012–) widely used in bioinformatics and beyond.

■ Pull-request review workflow

No direct commits to main branch without peer approval; forces explicit communication about what changed and why.

■ DVC (Data Version Control, 2019–)

Git-style versioning for large data files and ML pipelines; tracks data provenance alongside code.

P-HACKING / SELECTIVE REPORTING

Running many analyses and reporting only the significant ones; stopping data collection after reaching significance; covariate fishing; switching outcomes after seeing results. Produces a literature that overstates effect sizes and inflates false-positive rates.

SELF-REPORTED BY RESEARCHERS

**58% of psychologists
report ≥ 1 QRP**

The most common: continuing data collection after significant result in a subgroup; selectively reporting studies.

(John et al. 2012, Psych Science)

EVIDENCE OF SCOPE

01 John et al. (2012, Psych Science)

Survey of psychologists: 58% self-reported at least one questionable research practice; 22% reported continuing data collection after a significant subgroup result.

03 Masicampo & Lalande (2012)

Suspicious pileup of p-values just below .05 across psychology journals — probability near zero if results were genuine and unselected.

02 Simonsohn et al. (2014)

p-curve applied to JPSP studies where post-hoc covariate inclusion was suspected produced flat/left-skewed distribution — indicating lack of evidential value.

04 Head et al. (2015, PLOS Bio)

Excess just-significant p-values documented across many biological and social science fields — effect present even in top journals.

DETECTION TOOLS

01

p-curve (Simonsohn et al. 2014)

Distribution of significant p-values diagnoses evidential value vs. p-hacking; right-skew = real effects, left-skew / flat = p-hacking.

02

statcheck (Nuijten et al. 2016)

Automated extraction and recalculation of p-values; flags gross inconsistencies where the reported decision would change.

03

Caliper test (Gerber & Malhotra 2008)

Tests for overrepresentation of results just below the significance threshold vs. just above — a direct signature of selective stopping.

04

z-curve 2.0 (Bartoš & Schimmack 2022)

Estimates replication and discovery rates from published z-value distributions; quantifies the expected false-positive rate in a literature.

05

Registered vs. published audit

Compare pre-specified outcomes in trial registration or preregistration to outcomes actually reported in the paper.

TIER 2 A — TEAM LEVEL

P - HACKING / SELECTIVE REPORTING



INTRO



EVIDENCE



DETECTION



PREVENTION

PREVENTION & REMEDIATION

■ Preregistration (OSF 2013–)

Time-stamped, public registration of hypotheses and analysis plan before data collection; any deviation must be disclosed in the paper.

■ Registered Reports (2013–)

Peer review and provisional acceptance before data collection; journal commits to publish regardless of results if design is sound.

■ ClinicalTrials.gov (2000–)

Mandatory registration of clinical trial primary outcomes before unblinding; FDA requires results reporting within 12 months of completion.

■ AllTrials campaign (2013)

Advocacy for full public reporting of all trial results, positive and null; supports regulatory access to unpublished data.

HARKING

Hypothesising After Results are Known — presenting post-hoc hypotheses as if formulated before data collection. The introduction gets rewritten around what worked. Structurally indistinguishable from genuine prediction in the final paper.

SELF-REPORTED PREVALENCE

**~43% of researchers
have HARKed $\geq$ once**

Average across 6 survey studies. Many researchers do not recognise it as problematic because post-hoc theorising feels like a natural part of the scientific process.

EVIDENCE OF SCOPE

01 Kerr (1998)

Coined the term; theoretical analysis of 12 costs of HARKing for scientific progress. The core problem: HARKed hypotheses receive unwarranted evidential weight.

03 O'Boyle et al. (2017)

'Chrysalis effect': unconfirmed hypotheses are systematically removed between dissertation and published article — cleanly documented across matched pairs.

02 Review of 6 surveys (2017)

Average of ~43% of psychologists self-reported HARKing at least once; true rate likely higher due to social desirability bias.

04 OSC Reproducibility Project (2015)

Lower replication success correlated with large effect sizes and surprising results — both characteristics of HARKed findings.

DETECTION TOOLS

01

Registered vs. published comparison

Compare hypotheses in preregistration or clinical trial registration to those in the published paper; any new hypothesis is flagged.

02

Preprint timestamp audit

Compare posted preprint (with introduction) to final submission; hypothesis changes visible in diff.

03

No automated tool exists

Detection requires process-level evidence — preregistrations, lab records, email trails. This is why preregistration is the primary remedy.



PREVENTION & REMEDIATION

■ Preregistration (OSF 2013)

Time-stamped hypotheses make HARKing visible; any deviation from the registered plan must be disclosed as exploratory.

■ Transparent changes protocol (OSF)

Structured disclosure form for any deviations from the preregistered plan; distinguishes confirmatory from exploratory analyses in the published paper.

■ Registered Reports (2013–)

In-principle acceptance before data collection; journal commits to publish based on question and methods, not results — removes HARKing incentive entirely.

■ Exploratory / confirmatory distinction

Journals increasingly require authors to explicitly label each analysis as pre-specified or exploratory — normalising post-hoc theorising as legitimate but labelled.

FRAGILE FINDINGS: RESEARCHER DEGREES OF FREEDOM

Results that depend on arbitrary undisclosed analytic choices — which outliers to exclude, which covariates to include, how to operationalise variables. Technically honest but not robust. Gelman's 'garden of forking paths': researchers follow whichever path produces significance without conscious awareness of inflating false-positive rates.

SILBERZAHN ET AL. 2018

Effect size range:
0.89 to 2.93

29 teams, same soccer referee racial bias data.
Every team used valid methods. The spread reflects legitimate analytic freedom — not error, not fraud.

EVIDENCE OF SCOPE

01 Silberzahn et al. (2018, AMPPS)

29 independent teams, same data on soccer referee racial bias — effect size estimates ranged from 0.89 to 2.93 odds ratio. All used defensible methods.

03 Patel et al. (2015)

Vibration of effects in epidemiology: most published associations were highly sensitive to covariate choice — effect direction sometimes reversed.

02 Orben & Przybylski (2019)

Multiverse analysis on screen time and wellbeing: effects ranged from trivially negative to trivially positive across defensible specifications.

04 Simmons, Nelson & Simonsohn (2011)

'False-positive psychology': with legitimate-seeming flexibility, researchers can obtain a significant result for nearly any hypothesis.

DETECTION TOOLS

01

Multiverse analysis (Steege et al. 2016)

Runs all defensible analytic specifications and displays the distribution of results — makes fragility visible without accusing the researcher of misconduct.

02

Specification curve (Simonsohn et al. 2018)

Plots all results from valid specifications with statistical tests; the median estimate across the curve is less biased than the published single choice.

03

Vibration of effects (Patel et al. 2015)

Systematically varies covariate sets to show how unstable an epidemiological association is across equally justified model choices.

04

Pre-specified sensitivity analyses

Planned robustness checks committed to in advance; results must be reported regardless of whether they support the main finding.



PREVENTION & REMEDIATION

■ Pre-specified analysis plans

Committing to exact analytic choices — outlier rules, covariates, operationalisations — before data collection or unblinding removes post-hoc flexibility.

■ Open analysis code

Publishing full analysis code allows any reader to vary specifications and check robustness independently — the 'show your work' of quantitative science.

■ Robustness checks as field standard

Journals increasingly require at least one alternative specification; results that only survive one narrow choice are flagged as fragile.

■ FAIR data principles (2016)

Findable, Accessible, Interoperable, Reusable: if data and code are genuinely reusable, re-specification by anyone becomes practical.

PUBLICATION BIAS (TEAM DECISIONS)

Research teams selectively submit significant results and file-drawer null findings. At the team level this is a decision, not merely a field norm. Creates inflated effect sizes in published literature and undermines the value of uncorrected meta-analyses.

FRANCO ET AL. (2014, SCIENCE)

**68% of null results
never written up**

Among TESS government survey experiments with pre-registered designs — showing bias is not just a journal decision but a researcher decision made before submission.

EVIDENCE OF SCOPE

01 Sterling (1959)

First documented: ~97% of published psychology results were statistically significant — a probability near zero if this reflected the true rate of real effects.

03 Scheel, Schijen & Lakens (2021)

96% of psychology papers report at least one statistically significant primary finding — consistent with massive selective submission, not selective journal acceptance alone.

02 Franco, Malhotra & Simonovits (2014, Science)

68% of null results from TESS survey experiments (with pre-registered designs) were never written up, compared to 97% of strong positive results.

04 Fanelli (2010)

Proportion of papers reporting 'positive' or 'expected' results increased over 20 years across disciplines in Scopus; largest increases in applied fields.

DETECTION TOOLS

01	Funnel plot / Egger test (1997)	Detects asymmetry in meta-analytic effect size distributions indicating missing small-N null studies.
02	Trim-and-fill (Duval & Tweedie 2000)	Imputes missing studies and re-estimates corrected effect size; widely used in Cochrane reviews.
03	PET-PEESE (2008)	Precision-effect meta-regression: models effect size as a function of standard error to separate genuine effects from publication-bias inflation.
04	TES excess significance (Ioannidis 2007)	Compares expected to observed proportion of significant findings given estimated power; significant excess indicates file-drawer or p-hacking.

PREVENTION & REMEDIATION

■ Registered Reports (2013–)

Provisional acceptance before data collection removes post-hoc file-drawer decisions; paper is published regardless of results.

■ Journals of null results

European Journal of Negative Results, Journal of Negative Results in Biomedicine: dedicated venues that eliminate the bar against null findings.

■ PLOS ONE outcomes-neutral review (2006)

Accepts papers based on methodological quality alone, regardless of whether results are significant — first major journal to do so.

■ AllTrials / ClinicalTrials.gov

AllTrials (2013) advocates for full public reporting of all results; ClinicalTrials.gov mandates results reporting within 12 months of completion.

TIER 2B

**Group /
collusion-level
misconduct**

CITATION RINGS & UNCRITICAL PEER REVIEW

Can be intentional (orchestrated) or emergent (organic intellectual community). A prolific mentor's network of former students become the natural reviewers for each other's work — technically legitimate, but structurally likely to be uncritical. Hard to distinguish from a thriving research community without formal analysis.

SMALDINO & MCELREATH (2016)

Incentives naturally select for bad science

Evolutionary model: institutions that reward publication quantity over quality propagate low-powered, QRP-prone research practices — even without conscious intent.

EVIDENCE OF SCOPE

01 Smaldino & McElreath (2016)

Evolutionary model shows academic incentive structures naturally select for low-powered, QRP-laden research — 'the natural selection of bad science' even without fraud.

03 Ego-depletion & power posing literatures

Both fields show patterns consistent with uncritical peer review within a tight subfield community — followed by large pre-registered failures to replicate.

02 Ioannidis (2013)

Documented dense co-citation clusters in alternative medicine and some nutrition subfields; internal citations inflated apparent consensus and impact.

04 Chrysalis effect (O'Boyle et al. 2017)

Within tight research networks, manuscripts pass through multiple co-authors who share theoretical commitments, reducing adversarial review.

DETECTION TOOLS

01

Citation network analysis

Identifies unusually dense mutual-citation clusters relative to field baseline; network centrality measures reveal insular communities.

02

Reviewer history transparency

Open peer review makes reviewer identities and patterns visible over time; repeated pairing of same authors and reviewers becomes documentable.

03

Bibliometric anomaly detection

Flags journals with anomalously high self-citation or co-author citation rates relative to comparable journals in the same field.



PREVENTION & REMEDIATION

■ Double-blind peer review

Removes reviewer knowledge of author identity and institutional affiliation; reduces in-network bias in review decisions.

■ Cross-journal reviewer assignment

Drawing reviewers from outside an author's immediate network; some journals now require authors to suggest reviewers with no recent co-authorship.

■ Reviewer COI disclosure

Mandatory declaration of co-authorship, collaboration, and institutional relationship history; editors re-assign when conflicts exist.

■ Open peer review (F1000, eLife 2012–)

Published reviewer reports and reviewer identities allow post-publication scrutiny of review quality and conflicts of interest.

FAKE PEER REVIEW RINGS

Authors submit fabricated reviewer email addresses; paper mills arrange pre-agreed positive reviews; editors collude with author networks for rapid acceptance. Now accounts for roughly half of all retractions globally. Scale shifted from individual abuse to coordinated industrial operation.

RICHARDSON & AMARAL (PNAS 2025)

0.25% of editors
→ 30.2% of retractions

In PLOS ONE. The same concentrated pattern was found in Hindawi and IEEE proceedings — a structural signature of coordinated editorial fraud.

EVIDENCE OF SCOPE

01 Richardson & Amaral (PNAS 2025)

0.25% of PLOS ONE editors responsible for 30.2% of its retracted articles; network analysis shows editors routing submissions to each other — same pattern in Hindawi and IEEE.

03 Wiley / Hindawi (2022–24)

11,300+ papers retracted after systematic identification of coordinated fake review rings exploiting weak editorial workflows.

02 Scholarly Kitchen (2024)

Fake peer review now accounts for approximately half of all retractions globally — up from a rare occurrence a decade ago to the dominant retraction cause.

04 Cabanac & Labbé (2023)

Hundreds of editorial board members who accepted paper-mill products identified through systematic screening of retracted paper metadata.

DETECTION TOOLS

01	Editor network analysis (PNAS 2025)	Maps submission routing to identify editors who disproportionately handle each other's manuscripts and share retraction patterns.
02	Reviewer verification systems	Cross-check reviewer emails against institutional directories; flag free email addresses and unverifiable affiliations.
03	ORCID + institutional email (2012–)	Links reviewer identity to a verified institutional record; many publishers now require ORCID for all review assignments.
04	Statistical anomaly detection	Flags unusually fast turnaround, uniformly positive reviews, or repeated reviewer suggestions from the same submitting authors.

PREVENTION & REMEDIATION

■ Verified reviewer identity (ORCID)

ORCID (2012–) mandatory for review at most major publishers; links reviewer to a persistent scholarly identity verified against institutional records.

■ United2Act cross-publisher coordination

Shared intelligence on known bad actors across publisher networks; enables coordinated response when a ring spans multiple journals.

■ Publisher fraud detection (Wiley 2024)

Automated screening for coordinated review fraud patterns at the submission and assignment stage; shared across publisher networks.

■ Mass retraction protocols (COPE 2022–)

Standardised procedures for coordinated retraction of affected papers when a ring is identified; limits delays caused by case-by-case processing.

PAPER MILLS & PREDATORY JOURNALS

Industrial-scale fabrication: authorship for sale at \$500–\$3,000 per paper; APC-driven acceptance with no real peer review; journal hijacking using legitimate ISSNs. Richardson & Amaral (PNAS 2025): fraudulent papers may be doubling every ~1.5 years.

CANCER RESEARCH (AI SCAN 2024)

14.4% of molecular oncology papers in 2024

Flagged as potential paper-mill products — up from 0.06% in 1999. Patient harm is a direct consequence when fabricated clinical findings enter the evidence base.

EVIDENCE OF SCOPE

01 Richardson & Amaral (PNAS 2025)

Paper mill enabling networks 'large, resilient and growing rapidly'; fraud networks involve brokers, editors, and authors in coordinated schemes with resilient supply chains.

03 AI scan of cancer literature (2024)

14.4% of molecular oncology papers published in 2024 flagged as paper-mill products — up from 0.06% in 1999.

02 Problematic Paper Screener

32,786 suspected paper-mill products identified as of 2024 using tortured phrases, image clustering, and metadata anomalies.

04 Wiley / Hindawi (2022–24)

11,300+ papers retracted; 19 journals closed. Paper mill products continued to accumulate citations even after retraction — damaging the downstream literature.

DETECTION TOOLS

01

Problematic Paper Screener (2021)

Automated detection of tortured phrases, suspicious image clustering, metadata anomalies, and other paper-mill production signatures.

02

Tortured phrases detection

Identifies systematic obfuscation via machine translation — a production hallmark of mills that process manuscripts to evade plagiarism detectors.

03

Hijacked Journal Checker (RW)

Retraction Watch list (maintained by Anna Abalkina): 400+ hijacked journal ISSNs where scammer sites mimic legitimate journals.

04

Image duplication clustering (PNAS 2025)

Groups papers sharing image banks, identifying mill production batches; reveals coordinated fabrication at scale invisible to single-paper screening.



PREVENTION & REMEDIATION

■ DOAJ quality standards

Directory of Open Access Journals: maintains transparent criteria for legitimate OA publishers; being listed signals adherence to peer review and ethics standards.

■ United2Act cross-publisher coordination

Working group sharing mill identification intelligence across publishers; enables faster coordinated response than single-publisher action.

■ Think. Check. Submit. (2015)

Researcher guidance checklist for evaluating journal legitimacy before submission; widely adopted in library instruction for early-career researchers.

■ Indexing quality requirements

PubMed and Scopus delist journals that fail integrity standards; loss of indexing is commercially devastating and deters predatory operation.

INSTITUTIONAL MISCONDUCT COVER-UP

Universities slow-walking or suppressing misconduct investigations; protecting high-profile PIs at the expense of junior whistleblowers; ORI oversight capacity declining as the problem grows. The institution that benefits from grants has structural incentives to minimise findings.

ORI (2025)

Only 2 findings in 2025

Lowest since at least 2006. Average is ~10 per year. The office is partly hollowed out amid federal restructuring — at the same time the paper mill problem is accelerating.

EVIDENCE OF SCOPE

01 ORI (2025)

Only 2 misconduct findings released — fewest since 2006.
Average is ~10/year. Capacity has declined sharply amid federal restructuring of HHS.

03 Tessier-Lavigne (Stanford)

8-month formal investigation triggered only after sustained external pressure from science sleuths and media; final report found 'recklessness' rather than intent.

02 Hauser case (Harvard)

3-year internal investigation; Harvard confirmed misconduct but never disclosed details publicly — left the field unable to determine which of Hauser's hundreds of papers were tainted.

04 False Claims Act settlements

Private parties can sue on behalf of the federal government for fraudulent use of NIH/NSF funds; some of the largest misconduct cases settled this way without formal ORI finding.

DETECTION TOOLS

01

ORI oversight (capacity declining)

Federal investigators review institutional inquiry reports for NIH/HHS-funded research; declining capacity means fewer cases reach formal finding.

02

Investigative journalism

Science, Nature, Retraction Watch: external scrutiny routinely triggers institutional action that would not have occurred without public pressure.

03

False Claims Act litigation

Allows private parties to sue on behalf of government for fraud; some of the largest misconduct settlements reached via civil litigation, not ORI.



PREVENTION & REMEDIATION

■ ORI mandatory investigation timelines

Institutions required to complete inquiries within defined windows; persistent delays can trigger ORI's own investigation of the institution.

■ Federal whistleblower protections

Research Integrity Officer role at each institution; federal law prohibits retaliation against researchers who report misconduct to ORI.

■ Bik Science Integrity Fund (2025)

Financial and legal support for volunteer integrity advocates; launched in partnership with Retraction Watch / Center for Scientific Integrity.

■ Stockholm Declaration (2025)

Calls for independent institutional oversight bodies separate from university administration — removing the structural conflict of interest in self-investigation.

TIER 3

**Field /
community-level
problems**

PUBLICATION BIAS AT FIELD LEVEL

Journals preferentially publish significant, novel findings. Null results accumulate in file drawers. Effect sizes across entire literatures are inflated. The published record is not a representative sample of science conducted — it is a selection biased toward significance.

OSC REPRODUCIBILITY PROJECT (2015)

~36–50% of findings replicated

100 psychology studies directly replicated by independent teams. The most rigorous large-scale assessment of field-level publication bias consequences.

EVIDENCE OF SCOPE

01 OSC Reproducibility Project (2015, Science)

~36–50% of 100 psychology findings replicated when directly attempted by independent teams. Effect sizes in replications were on average half the size of originals.

03 z-curve meta-analyses (2022)

Estimated replication rate ~50–60% across social sciences when correcting for publication bias in z-value distributions.

02 Camerer et al. (2018, Nature Human Behaviour)

62% replication rate in economics laboratory experiments; original effect sizes approximately twice the replication effect sizes.

04 Ioannidis (2005)

Theoretical model: 'Why most published research findings are false' — when power is low, bias is present, and many hypotheses are tested, most significant findings are false positives.

DETECTION TOOLS

01	p-curve across literatures (2014)	Diagnoses whether a body of literature has evidential value — flat or left-skewed distribution indicates absence of true effects or intense p-hacking.
02	z-curve replication rate estimates (2022)	Estimates expected replication success rate from published z-values; quantifies how much of a literature is likely false positive.
03	TES excess significance (Ioannidis 2007)	Compares expected to observed proportion of significant results given estimated power; significant excess implies file-drawer or p-hacking at field scale.
04	Bias-corrected meta-analysis	PET-PEESE and robust Bayesian meta-analysis adjust for selection bias; provide less inflated effect size estimates for use in power calculations.

PREVENTION & REMEDIATION

■ Registered Reports format (2013–)

Peer review and acceptance before data collection; paper published regardless of results. Cleanest structural fix for field-level publication bias.

■ Replication journals (AMPPS 2018)

Advances in Methods and Practices in Psychological Science: dedicated venue for pre-registered replications; Royal Society Open Science has a parallel track.

■ PLOS ONE outcomes-neutral review (2006)

First major journal to accept based on methodological quality alone; model adopted by hundreds of journals since.

■ Pre-results peer review

Editorial commitment before data collection or unblinding: the journal evaluates the question and design, not the outcome — eliminates incentive to selectively report.

PUBLISH-OR-PERISH & APC ECONOMICS

Tenure and promotion tied to publication count and impact factor. NIH R01 success rates have declined from ~31% (2000) to ~19% (2024). APC-driven open access rewards volume over quality. Stapel (2011): 'The pressure became too much... I wanted too much, too fast.'

NIH R01 SUCCESS RATES

~31% (2000)
→ ~19% (2024)

Historic decline in grant success rates intensifies competition and the pressure to produce significant publishable results at every stage of a research career.

EVIDENCE OF SCOPE

01 John et al. (2012, Psych Science)

Career pressure correlated with self-reported QRPs in psychology survey; researchers at early career stages reported highest QRP rates.

03 Elsevier profit: ~\$3.6B (2023)

Wiley/Hindawi collected APC revenue even for fraudulent paper-mill products before mass retraction — APC model creates incentive to accept regardless of quality.

02 NIH historical success rate data

R01 grant success rates fell from ~31% (2000) to ~19% (2024); the fraction of funded researchers is at a historic low.

04 Fanelli (2010)

Proportion of papers reporting 'positive' or 'expected' results increased over 20 years across disciplines; largest increases in applied and clinical fields.

DETECTION TOOLS

01

Bibliometric audits

Measure publication count, citation inflation, and h-index gaming by institution, country, or field; track how incentive changes affect output patterns.

02

QRP prevalence surveys

Periodic surveys benchmark self-reported questionable practices; John et al. (2012) model has been replicated across disciplines and countries.

03

Retraction rate by institution

Comparing institutional retraction rates to output identifies systematic problems; Retraction Watch database enables this analysis.



PREVENTION & REMEDIATION

■ DORA declaration (San Francisco 2012)

Signatories commit to not using journal impact factor in hiring, promotion, or funding decisions; now has thousands of institutional signatories.

■ Stockholm Declaration (2025)

34 specific actions including abandoning publication counting in hiring and tenure decisions; calls for structural reform of research assessment globally.

■ Leiden Manifesto (2015)

10 principles for responsible use of quantitative metrics in research evaluation; counters over-reliance on publication count and JIF.

■ Diamond OA / subsidy models

Decouples access from APC revenue; removes the financial incentive to accept papers regardless of quality; funded by institutions or funders directly.

LACK OF REPLICATION CULTURE

No career credit for replication studies. Novel findings rewarded over verification. Failed replications difficult to publish in top journals.

Fraud-detection work by volunteer sleuths receives no formal academic credit despite enormous public value.

MUELLER-LANGER ET AL. (2019)

**< 0.1% of top
economics papers
were replications**

Among top 50 economics journals 1974–2014.
Psychology is modestly better — ~1% of top
100 journal articles since 1900 qualify as
replications.

EVIDENCE OF SCOPE

01 **Mueller-Langer et al. (2019)**

Fewer than 0.1% of articles published in top 50 economics journals from 1974 to 2014 were replications — an almost complete absence of verification.

03 **Many Labs 1–4 (2013–2018)**

Large variation in replication success; original effect size is the most reliable predictor of replication outcome — large effects replicate, small ones often do not.

02 **Makel, Plucker & Hegarty (2012)**

Approximately 1% of articles in top 100 psychology journals since 1900 qualify as replications; the rate has improved since 2011 but remains very low.

04 **No credit for sleuthing**

Elisabeth Bik left a stable industry position to do integrity work full-time — financially precarious and professionally unrecognised by standard academic metrics.

DETECTION TOOLS

01

OSC Reproducibility Project (2015)

First large-scale systematic assessment; 100 psychology studies replicated across dozens of independent teams — established the benchmark for field-level replication work.

02

Many Labs projects (2013–2018)

Multi-site coordinated replications enabling detection of context dependence vs. genuine failure to replicate; Many Labs 2 tested 28 effects in 36 countries.

03

Institute for Replication (i4rep, 2022–)

Independent infrastructure for economics and social science replication; connects replicators with original authors and journals for efficient verification.

TIER 3 — FIELD / COMMUNITY LEVEL

LACK OF REPLICATION CULTURE



INTRO



EVIDENCE



DETECTION



PREVENTION

PREVENTION & REMEDIATION

■ CRediT taxonomy (CASRAI 2014)

14 contributor roles including 'Validation' — enables replication credit in the author byline; standardised across thousands of journals.

■ Sleuths in Residence (RW, 2024)

Paid positions for science integrity advocates at Retraction Watch; formal recognition and support for a role that was entirely volunteer-based.

■ Replication journals & tracks

Advances in Methods and Practices in Psychological Science (AMPPS, 2018); Royal Society Open Science replication track; Journal of Negative Results.

■ Big Team Science / PSA (2017–)

Psychological Science Accelerator: large multi-site collaborations that build replication into initial study design; Many Labs format institutionalised.

POOR DATA STEWARDSHIP & INACCESSIBLE RESEARCH

Data lost after publication through staff turnover, institutional change, or format obsolescence. Code not shared or not executable. No persistent identifiers. Barriers to auditing allow errors and fraud to persist undetected for years after publication.

VINES ET AL. (2014)

Data availability declines ~17% per year

After publication. A paper 20 years old has a near-zero probability of yielding recoverable data on request — regardless of any stated availability.

EVIDENCE OF SCOPE

01 Stodden, Seiler & Ma (2018, Science)

Contacted authors of 204 Science papers — only 26% shared data and code on request, despite the journal's data policy requiring it.

03 Vines et al. (2014)

Data availability declines ~17% per year after publication — a paper from 20 years ago has virtually no recoverable data, regardless of its stated availability.

02 Gabelica et al. (2022)

3,556 biomedical papers with data availability statements: 79% of the data was unavailable when actually requested from the authors.

04 Eklund et al. (2016, PNAS)

fMRI software bug was discoverable only because researchers eventually obtained access to the original processing pipeline — data access enabled the correction.

DETECTION TOOLS

01

Computational reproducibility checks

Independent researchers attempt to re-run published code on shared data; the gap between 'data available' and 'data actually usable' is diagnostic.

02

Data availability statement audits

Track compliance between stated and actual data access; Gabelica et al. (2022) documented the gap at 79% unavailability despite stated availability.

03

FAIR compliance assessment tools (2021–)

Automated scoring of datasets against Findability, Accessibility, Interoperability, Reusability criteria; FAIRshake and F-UJI used by funders.



PREVENTION & REMEDIATION

■ FAIR data principles (2016)

Wilkinson et al.: Findable, Accessible, Interoperable, Reusable — the foundational framework now adopted by NIH, NSF, Wellcome, and most major funders.

■ Zenodo / Figshare / Dryad / OSF (2012–)

Free infrastructure for depositing research data and code; institutional repositories supplemented by these discipline-agnostic platforms.

■ RO-Crate packaging (2019–)

Container format preserving data, code, metadata, and provenance information together — the research object as a citable, shareable, executable unit.

■ NIH data sharing mandate (2023)

All NIH-funded research must deposit data in a public repository meeting FAIR standards; largest single mandate in research data history.

■ Data DOIs & versioning (DataCite 2009–)

Persistent identifiers for datasets; version tracking ensures the exact dataset used in a paper is permanently citable and distinguishable from later versions.

REVIEWER BURNOUT & DECLINING PEER REVIEW QUALITY

Volume of manuscript submissions far exceeds reviewer capacity. AI-generated submissions are accelerating the problem. Superficial review allows errors and fraud through. The system relies on uncompensated expert labour with no formal career credit mechanism.

BRUBAKER (2023)

**~15 million review
hours wasted annually**

Estimated on papers that are desk-rejected without being reviewed — reviewer time consumed before any editorial decision, with zero scientific benefit.

EVIDENCE OF SCOPE

01 **Publons Global State of Peer Review (2018)**

Reviewer workload growing faster than the research community; declining willingness to review; ~15M review requests per year globally estimated.

02 **Brubaker (2023)**

Estimated ~15 million peer review hours wasted annually on papers that are desk-rejected without review — before any editorial decision is made.

03 **Nature (2023–)**

Editors report dramatic increases in AI-generated manuscript submissions, stretching review capacity and requiring new detection workflows at submission.

04 **Mass resignation events (2022–23)**

40+ scientists resigned from Neuroimage (Elsevier) editorial board over excessive APC fees; entire editorial board of Journal of Human Evolution resigned (Dec 2024).

DETECTION TOOLS

01

Open peer review audit trails

Published reviewer reports allow post-publication assessment of review quality and depth; accountability through transparency.

02

Post-publication review (PubPeer 2012–)

Crowd-sourced error detection compensates for pre-publication review failures; has triggered retractions at rates journals alone would not achieve.

03

Review quality scoring (proposed)

Rubric-based scoring of reviewer thoroughness — no widely adopted standard yet; Publons and Web of Science track reviewer output, not quality.



PREVENTION & REMEDIATION

■ Open peer review (eLife, F1000 2012–)

Reviewer reports published alongside paper; reviewer accountability through transparency; reviewers credited by name for their intellectual contribution.

■ Portable peer review (Review Commons 2019)

A single peer review transfers across multiple journals; eliminates duplicated reviewer effort when authors resubmit after rejection.

■ Reviewer recognition (Publons 2012)

Formal attribution of peer review work to researcher profiles; now integrated into Web of Science; recognition visible in hiring and promotion processes.

■ AI-assisted pre-screening (2023–)

Automated desk rejection of clearly out-of-scope or AI-generated manuscripts frees reviewer time for genuine papers; deployed by Nature, Springer Nature, and others.